

RUSTOM FIRDOSH BHESANIA

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EDUCATION

Relevant Coursework: Digital Signal Processing, Deep Learning for Speech & Text Processing, Microphone Array Processing, Experimental Phonetics, Speech Enhancement and Speech Analysis.

Master of Science in Computational Linguistics, University of Stuttgart, Germany Oct 2024 - Current

Bachelor of Technology in Electronics and Telecommunication Engineering

SVKM's NMIMS UNIVERSITY- Mukesh Patel School of Technology Management and Engineering, Mumbai, India Jul 2016 - Aug 2022

ACADEMIC RESEARCH PROJECT

Universal Test-Time Adaptation for Speech Enhancement Dec 2025 - Aug 2026

Institute for Natural Language Processing (IMS) in collaboration with Signal Processing and System Theory (ISS), University of Stuttgart

Supervisor: Tobias Raichle; Examiner: Dr. Wolfgang Wokurek (IMS, Digital Phonetics)

- Studied whether **test-time adaptation (TTA)** methods built for denoising transfer to dereverberation, training four **CMGAN** models (noise-only, reverb-only, and two jointly trained on both) and evaluating four TTA methods, including RIRmixIT, a reverb-specific method developed for the project.
- Showed that existing TTA methods (MPol, LaDen, RemixIT) all degrade performance on reverberation, since their adaptation assumptions hold for additive noise but not convolutive distortion; updates concentrate in the shared conformer blocks' normalisation statistics, useful only after joint training.
- Central result: adapting a jointly-trained model on one distortion and evaluating on the other, with no adaptation on the second, gives substantially larger gains than adapting directly on the target distortion.

PROJECTS

WorldSounds (Team Sundarsheng): Cross-Tradition Music Intelligence. Built at the [MTG Generative Music AI Workshop](#) (UPF, Barcelona): a knowledge base of Carnatic and Chinese Jingju features (CREPE pitch, gamak/meend scores, pentatonic adherence) driving a 2nd-order Markov chain melody generator. Jul 2026

Karaoke Studio AI-Powered Music Source Separation. Built a full-stack karaoke app for the Speech Analysis course (Dr. Wolfgang Wokurek) that separates vocals from instrumentals in real time using Meta's Demucs v4. Pipeline: YouTube URL; yt-dlp download; Demucs separation; synced lyrics via React frontend with dual volume control. Flask backend; 9.20 dB SDR. [\[GitHub\]](#) 2026

Room Acoustics Signal Propagation Study. Investigated room acoustic principles using a Respeaker Mic Array 2.0 and Python's pyroomacoustics library to model signal propagation. 2025

Comparative Phonetic Study of Hindi and Awadhi Languages. Traced key phonetic changes across Hindi and Awadhi languages within their shared historical context. 2025

Music Learning System with DTW and Chromagrams. Built a system giving objective feedback by comparing student and teacher recordings via Chromagrams (melody) and DTW (rhythm). [\[GitHub\]](#) 2025

Audio Deepfake Detection. Investigated ASVspoof 2019 to identify optimal acoustic features for detecting synthetic speech; presented findings in a poster session on feature effectiveness across TTS systems. [\[GitHub\]](#) 2025

SKILLS

Technical Skills	Python, Librosa, music21, Opencv, NLTK, pyroomacoustics
Web & API Frameworks	Fast-API, Flask, Django
Deep Learning Frameworks	Pytorch, Keras
Music Source Separation	Demucs, Spleeter
ASR Tool-kits	Kaldi, pykaldi, pytorch-kaldi, Espnet, Speech Brain
Open Source Tool-kits	Linux, Git, Docker, Bash Scripting

EXPERIENCE

Student Assistant, Visualization Research Center (VISUS)

Sedlmair Group, University of Stuttgart

May 2026 - Sep 2026

Stuttgart, Germany

- Built **ViSuS**, a rehearsal-recording comparison tool with Simeon Rau (Sedlmair Group): librosa features (tempo, key, timbre, harmony, rhythm), **DTW** alignment, MIDI score alignment, and pairwise similarity scoring. [\[GitHub\]](#)
- Designed a **FastAPI** backend serving two Streamlit clients (full research view and a plain-language view piloted with a university band), with similarity matrix, stream graph, and MDS recording map.

Student Assistant, Digital Phonetics Lab

Institute of Natural Language Processing (IMS), University of Stuttgart

Apr 2025 - Current

Stuttgart, Germany

- Managed professional audio equipment (Yamaha digital mixer, microphones) and the full recording chain.
- Conducted **high-fidelity audio and music data acquisition** in a professional studio, creating robust datasets for phonetic and NLP research.

Research Intern

K.L University, Under Guidance of Dr. K. Samudravijaya

Jan 2022 - Apr 2022

Andhra Pradesh, India

- Explored **ASR toolkits** (Pytorch-kaldi, Espnet, Speech Brain) and Wav2Vec, and developed **End-to-End ASR Systems** on English, Hindi, Marathi, and Gujarati data.

Research Intern

Indian Institute of Technology Dharwad, Under Guidance of Dr. K. Samudravijaya

Jul 2021 - Dec 2021

Dharwad, India

- Gained deep expertise in **Automatic Speech Processing using the Kaldi tool-kit**, building ASR systems for Assamese, Hindi, English and a **custom Parsi (Gujarati) digit database**.

ENTREPRENEURSHIP

Cyber Valley AI Incubator (Batch 9)

Max Planck Institute for Intelligent Systems

5 Oct 2026 - 16 Oct 2026

Stuttgart, Germany

- Building **Voclair**, a local-first speech enhancement toolkit that removes noise and reverberation from microphone audio with very low delay, giving people in a room and AI agents clean speech. Runs on-device with no cloud; ships as an SDK for developers and live sound users.
- Pitching at Demo Day (16 Oct) to investors and industry experts after workshops and founder mentoring.

Corporate Campus Challenge (7th Batch), Campus Founders

Challenge Partner: Scheer IMC (imc Learning Suite)

Apr 2026 - Jul 2026

Heilbronn, Germany

- Applied **Design Thinking** and the **Double Diamond** process, including personas and user interviews, to a real challenge from Scheer IMC (imc Learning Suite).
- Proposed **peer-to-peer learning** and an AI-driven **course recommendation system** to improve employee engagement on the LMS.
- Presented a working prototype to stakeholders at the program's final demo day.

AWAKE: From Idea to Startup in a Few Weeks, ENI, University of Stuttgart

Apr 2026 - Jul 2026

- Applied customer discovery, persona building, and value proposition design (ENI) to a karaoke source-separation pitch, which evolved into an **“anywhere karaoke” system** turning two phones into wireless mics paired to a laptop display for instrument-free, real-time scored karaoke.
- Presented the refined concept and business model at the program's pitch finale.

WORKSHOPS, ACTIVITIES AND INTERESTS

Assistant Sound Engineer (Voluntary)

Everybody Speaks Music, University Band, Stuttgart, Germany

Apr 2025 - Current

- Provide technical support for live performances, including stage setup, rigging, and troubleshooting.
- Operate digital audio mixers to manage input levels and conduct microphone checks, applying theoretical knowledge of audio signals and acoustics in a live sound reinforcement setting.

Personal Interests: Speech and Music Processing, exploring how musical traditions originated; **Linux Enthusiast**, following distributions, file systems, CLI workflows, and scripting.